Simple Classification Model Using Decision Tree on Myntra T-Shirt Dataset

1. Introduction

The objective of this project is to build a simple classification model using the Myntra T-Shirt Dataset. A Decision Tree Classifier is used to classify T-shirts into two categories: High Discount and Low Discount. The model is evaluated using Accuracy, Precision, and Recall metrics.

2. Dataset Description

The dataset was obtained from Kaggle and contains information about T-shirt products listed on Myntra.

Attributes Used

- Brand
- Title
- Selling Price
- Price
- Discount

Total Records

- 4950 rows
- 5 columns

3. Data Preprocessing

Missing Values

The dataset was checked for missing values using:

df.isnull().sum()

Observation: No missing values were found.

Duplicate Records

Duplicate records were identified and removed using:

df.drop_duplicates(inplace=True)

Data Type Conversion

The Price and Discount columns were converted into numerical format for analysis and model training.

df['Price'] = pd.to_numeric(...)

df['Discount'] = pd.to_numeric(...)

Target Variable Creation

A new target column named **High_Discount** was created.

df['High_Discount'] = (df['Discount'] > 40).astype(int)

Where:

- 1 = High Discount
- 0 = Low Discount

4. Data Splitting

The dataset was divided into Training and Testing sets.

- Training Data: 80%
- Testing Data: 20%

```
train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

5. Classification Model

A Decision Tree Classifier was used to train the model.

from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

Decision Tree was selected because it is simple, efficient, and easy to interpret.

6. Model Evaluation

The trained model was evaluated using:

Accuracy

Measures how many predictions were correct.

accuracy_score(y_test, y_pred)

Precision

Measures the proportion of positive predictions that were actually positive.

precision_score(y_test, y_pred)

Recall

Measures how many actual positive cases were correctly identified.

recall_score(y_test, y_pred)

Classification Report

classification_report(y_test, y_pred)

Result Screenshot:

RESULTS

Accuracy : 0.997979797979798

Precision: 0.9975490196078431

Recall : 0.9975490196078431

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	582
1	1.00	1.00	1.00	408
accuracy			1.00	990
macro avg	1.00	1.00	1.00	990
weighted avg	1.00	1.00	1.00	990

7. Results

The Decision Tree model successfully classified products into High Discount and Low Discount categories.

Performance Metrics

Metric	Value
Accuracy	Accuracy : 0.997979797979798
Precision	Precision: 0.9975490196078431
Recall	Recall : 0.9975490196078431

8. Conclusion

A Decision Tree Classification model was successfully developed using the Myntra T-Shirt Dataset. The dataset was preprocessed, split into training and testing datasets, and used to train the model. The performance was evaluated using Accuracy, Precision, and Recall metrics. The model demonstrated the ability to classify T-shirt products based on discount levels and can be further improved using advanced machine learning techniques.